UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF M.Sci. WITH HONOURS

FOURTH-YEAR FINAL EXAMINATION

03 23560

LM Inference from Scientific Data

SEMESTER 1 EXAMINATIONS 2021/22

Time Allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

All symbols have their usual meanings.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and a table of physical constants and units that may be required will be found at the end of this question paper.

Answer **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

1. (i) State the definitions given in lectures for the moment generating function (MGF) $M_x(t)$ and the central moments μ_k of a continuous probability distribution. **[2]**

(ii) Explain why the MGF is useful for computing the central moments and give an expression for μ_k in terms of $M_x(t)$. **[2]**

A random variable x follows a χ^2 distribution with $\nu > 0$ degrees of freedom; the probability density function for this distribution is given by

$$f_\nu(x) = \begin{cases} \dfrac{x^{(\nu-2)/2} \exp\left(-x/2\right)}{2^{\nu/2}\Gamma\left(\nu/2\right)} & \text{if } \ x > 0 \\ 0 & \text{otherwise} \end{cases} .$$

(iii) Find the MGF $M_x(t)$ (for $t < 1/2$) for the χ^2 distribution $f_\nu(x)$. You may use the following standard integral which holds for any $\alpha > -1$, **[8]**

$$\int_0^\infty \mathrm{d}x \; x^\alpha \exp(-x) = \Gamma(1+\alpha) \; .$$

(iv) Hence or otherwise, find the mean and variance of the χ^2 distribution. **[10]**

(v) Sketch on the same set of axes at least two versions of the χ^2 distribution with different values of ν. **[4]**

(vi) Find the mode of the χ^2 distribution (i.e. the value of x at which $f_\nu(x)$ is maximum). Hint: consider the cases $\nu < 2$ and $\nu > 2$ separately. **[4]**

2. The value of a quantity X is measured N times and the following values obtained:

$$\{x_1, x_2, \ldots, x_N\}\,.$$

The errors on each measurement are assumed to be independent and Gaussian distributed with a known standard deviation of σ_i for the i^{th} measurement. (Note the standard deviations are different for each measurement.)

(i) Write down an expression for the likelihood $P(\{x_i\}|X)$. **[2]**

(ii) Using a Gaussian prior $P(X)$ (also known as a normal prior) with a mean value of μ and a standard deviation of Δ, write down Bayes' theorem for this problem and an expression for the posterior $P(X|\{x_i\})$. **[6]**

(iii) Find the maximum *a posteriori* value (i.e. the most likely) of X. **[5]**

(iv) Define the Bayesian evidence, Z, and compute Z for this problem **[6]**

Now suppose that instead of a flat prior we instead used Jeffreys' prior on the parameter X between X_{min} and X_{max}.

(v) Write down a correctly normalised expression for the new Jeffreys' prior $P(X)$ and sketch both the Jeffreys' prior and the Gaussian prior from the earlier parts of the question on the same set of axes. **[3]**

(vi) Write down a new expression for the posterior $P(X|\{x_i\})$ using Jeffreys' instead of the Gaussian prior used in part (ii) of the question. (You do *not* need to normalise this posterior.) **[2]**

(vii) Using the new posterior that you found in part (vi), show that the new maximum posterior value $\hat{X}$ must satisfy the following quadratic equation, **[6]**

$$\hat{X}^2 \sum_i \left(\frac{1}{\sigma_i^2}\right) - \hat{X} \sum_i \left(\frac{x_i}{\sigma_i^2}\right) + 1 = 0\,.$$

3. A two-dimensional probability distribution has probability density function (PDF) $P(x, y)$.

   (i) State the definitions given in lectures of the covariance $\mathrm{Cov}(x, y)$ and the correlation $\rho(x, y)$ in terms of the PDF $P(x, y)$ clearly defining any other quantities that you use in your definition. **[5]**

   Consider the probability distribution

$$P(x,y) = \begin{cases} C\left[\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2\right] & \text{if } 0 < x < a \text{ and } 0 < y < b \\ 0 \quad \text{otherwise}\,. \end{cases}$$

   where C is a normalisation constant.

   (ii) Sketch a plot of the distribution $P(x, y)$ with x and y on the horizontal axes and contours indicating the value of $P(x, y)$. **[2]**

   (iii) Compute the covariance and the correlation for this distribution. **[8]**

   Suppose now that we have N samples where each data point consists of a pair of numbers:

$$\{(x_i, y_i) \text{ for } i = 1, 2, \ldots, N\}\,.$$

   (iv) State the definitions of the sample covariance $V_{x,y}$ and the sample correlation $r_{x,y}$ in terms of this data. **[5]**

   Suppose we measure the following data with $N = 5$:

$$\{(1,1), (2,3), (1,2), (4,3), (3,2)\}\,.$$

   (v) Compute the sample covariance $V_{x,y}$ and the sample correlation $r_{x,y}$ for this data. **[3]**

   (vi) Compute the Pearson correlation t_p statistic which is defined as **[2]**

$$t_p = \frac{r_{x,y}\sqrt{N-2}}{1 - r_{x,y}^2}\,.$$

   (vii) Explain how you would use the Pearson correlation statistic to perform a quantitative test for correlation in the data including details of any assumptions needed for the test. **[5]**

Formula Sheet

Any formulae that you may need are given in the individual questions.

Physical Constants and Units

Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	$273.15\,\mathrm{K}$
Avogadro constant	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\,\mathrm{m^{-1}}$
	$R_\infty hc$	$13.606\,\mathrm{eV}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
	$\hbar c$	$197.3\,\mathrm{MeV\,fm}$
Charge of proton	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Mass of electron	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Rest energy of electron		$0.511\,\mathrm{MeV}$
Mass of proton	m_p	$1.673 \times 10^{-27}\,\mathrm{kg}$
Rest energy of proton		$938.3\,\mathrm{MeV}$
One atomic mass unit	u	$1.66 \times 10^{-27}\,\mathrm{kg}$
Atomic mass unit energy equivalent		$931.5\,\mathrm{MeV}$
Electric constant	ϵ_0	$8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	$7.297 \times 10^{-3} = 1/137.0$
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12}\,\mathrm{m}$
Bohr radius	a_0	$5.2918 \times 10^{-11}\,\mathrm{m}$
angstrom	Å	$10^{-10}\,\mathrm{m}$
barn	b	$10^{-28}\,\mathrm{m^2}$
torr (mm Hg at 0 °C)	torr	$133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$)